1415. The k-th Lexicographical String of All Happy Strings of Length n

A happy string is a string that:

consists only of letters of the set ['a', 'b', 'c'].

s[i] != s[i + 1] for all values of i from 1 to s.length - 1 (string is 1-indexed).

For example, strings "abc", "ac", "b" and "abcbabcbcb" are all happy strings and strings "aa", "baa" and "ababbc" are not happy strings.

Given two integers n and k, consider a list of all happy strings of length n sorted in lexicographical order.

Return the kth string of this list or return an empty string if there are less than k happy strings of length n.

Example 1:

Input: n = 1, k = 3

Output: "c"

Explanation: The list ["a", "b", "c"] contains all happy strings of length 1. The third string is "c".

Example 2:

Input: n = 1, k = 4

Output: ""

Explanation: There are only 3 happy strings of length 1.

Example 3:

Input: n = 3, k = 9

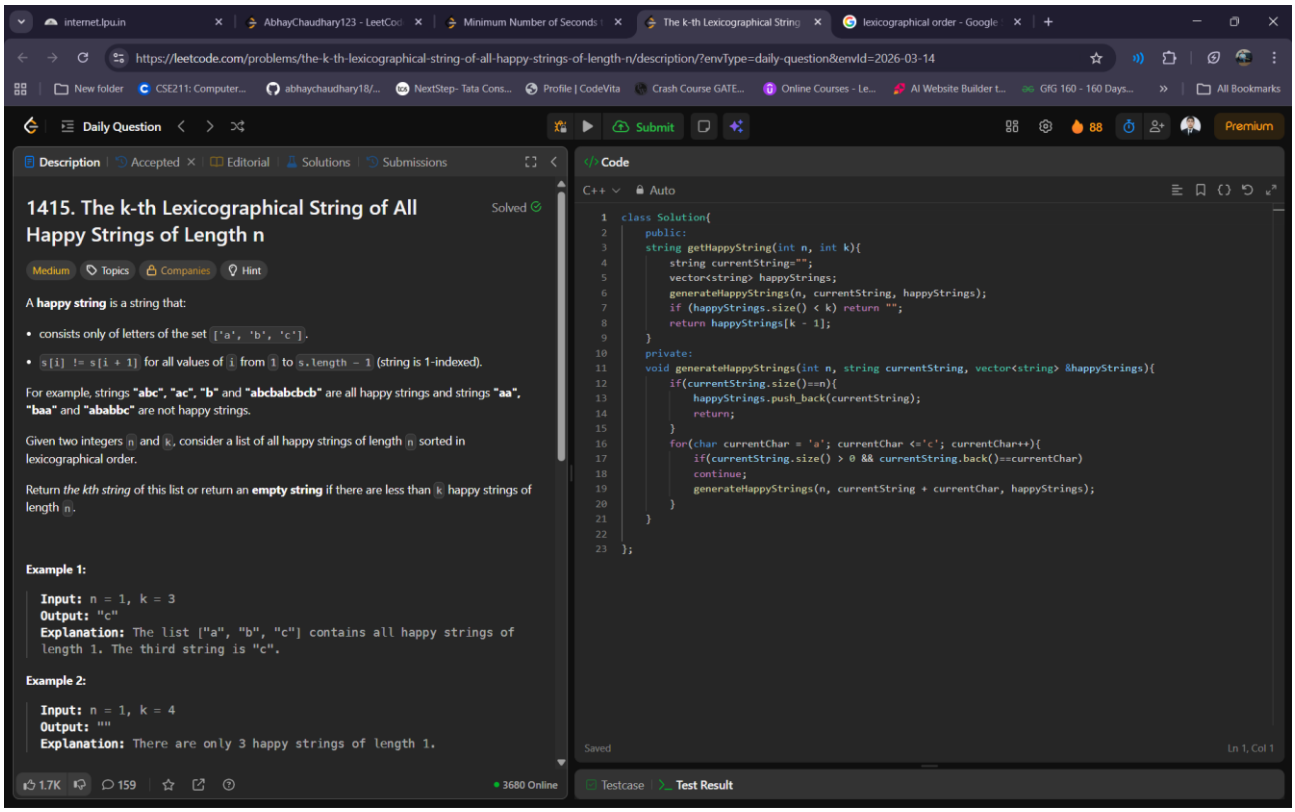
Output: "cab"

Explanation: There are 12 different happy string of length 3 ["aba", "abc", "aca", "acb", "bab", "bac", "bca", "bcb", "cab", "cac", "cba", "cbc"]. You will find the 9th string = "cab"

Constraints:

1 <= n <= 10

1 <= k <= 100



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Solved

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- consists only of letters of the set `['a', 'b', 'c']`.
- `s[i] != s[i + 1]` for all values of `i` from `1` to `s.length - 1` (string is 1-indexed).

For example, strings **"abc", "ac", "b"** and **"abcbabcbcb"** are all happy strings and strings **"aa", "baa"** and **"ababbc"** are not happy strings.

Given two integers `n` and `k`, consider a list of all happy strings of length `n` sorted in lexicographical order.

Return *the kth string* of this list or return an **empty string** if there are less than `k` happy strings of length `n`.

Example 1:

```
Input: n = 1, k = 3
Output: "c"
Explanation: The list ["a", "b", "c"] contains all happy strings of length 1. The third string is "c".
```

Example 2:

```
Input: n = 1, k = 4
Output: ""
Explanation: There are only 3 happy strings of length 1.
```

```cpp
class Solution{
    public:
    string getHappyString(int n, int k){
        string currentString="";
        vector<string> happyStrings;
        generateHappyStrings(n, currentString, happyStrings);
        if (happyStrings.size() < k) return "";
        return happyStrings[k - 1];
    }
    private:
    void generateHappyStrings(int n, string currentString, vector<string> &happyStrings){
        if(currentString.size()==n){
            happyStrings.push_back(currentString);
            return;
        }
        for(char currentChar = 'a'; currentChar <='c'; currentChar++){
            if(currentString.size() > 0 && currentString.back()==currentChar)
            continue;
            generateHappyStrings(n, currentString + currentChar, happyStrings);
        }
    }

};
```